05506026

Digital Image Processing

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Computer Science, KMITL

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A bit about me…

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BSc Applied Mathematics – KMITL
MSc Computer Games Technology – University of Abertay Dundee (Scotland)
PhD Computer Science – The Mixed Reality Lab, University of Nottingham (England)
Face-to-face interactions in augmented reality games

A game researcher, game developer, game designer, game analyser, game player, and game addict.

Research interest: Computer games, Exertion games, Augmented reality, Mixed reality, Virtual reality, Everything about mixing reality with life in entertainment domain especially – of course – games

Where to find me?

Room Sc8 – 701

Check my availabilities on Google Calendar (Normally Thursday): **https://goo.gl/AfRDzt**

Out of that time, allocate a slot with me at **Witchaya.To@kmitl.ac.th** or in Google Classroom system

CONTENTS

COURSE OVERVIEW

What is this course about?

- This course will help you to build a foundation of digital image processing theories and techniques
- The course is a combination between mathematics and coding

Course Outline

13 times of lecture

All Materials are in English

2 Times a week
(Will end at midterm)

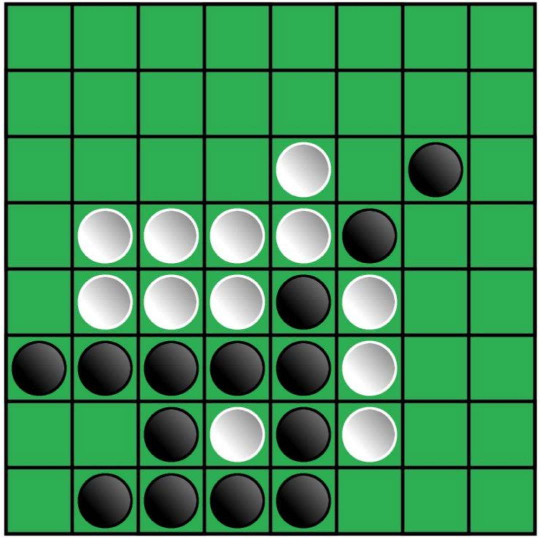
Game-styled, Grades are solely on the game

You are going to play a game across this semester

- You are all now a level 1 player and try to maximise your level along the journey
- The max level is 20
- Your EXP will be earned by
  - Complete quests (the quest will be in lab sheet)
  - Playing game
- Your final grade will be 100% determined from your final level

You are going to play a game across this semester

- The game is simple…
- Otello!



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1															
2															
3															
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5															
6															
7															
8															
9															
10															
11															
12															
13															
14															
15															

You are going to play a game across this semester

1. Complete a quest (or any special occasions such as answer a question in class), you have right to put your avatar in the board (sometimes 1, sometimes more than that)
   - If you don’t have any avatar in the board, you can put only in a green square
   - If you’ve already had an avatar, you can put your avatar next to your previous ones.

You are going to play a game across this semester

2. If your avatar is next to another, you can put your new avatar next to the others and conquer in between

You are going to play a game across this semester

3. At midnight of each Thursday, I will give you 1 EXP per square you conquer (10 EXP at most)
4. At the beginning of Friday class, I will random a square (fairness guaranteed by www.random.org) to give 20 EXP to the player who conquer that square (if no one on that square, no EXP will be given)
5. You can finish the quest any time you want before the ending of the game *(plan your strategy wisely)*
6. The game ends at **midnight of Sunday 7th September 2025**

Now, sign up your avatar

https://forms.gle/F2x9SSZt4HGjWz6cA

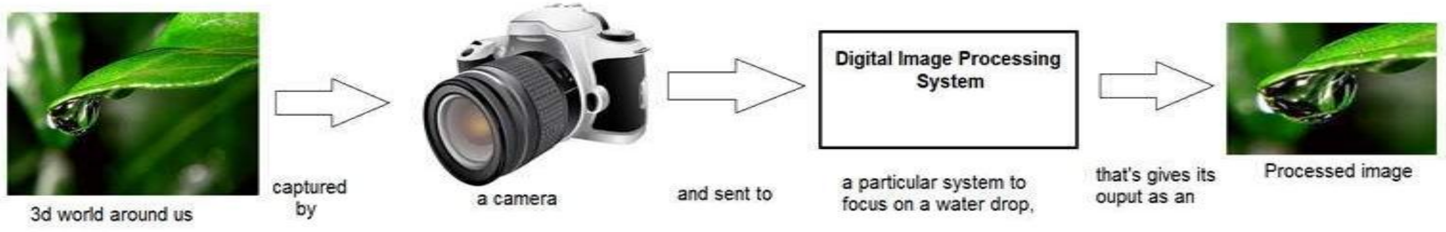
Your first quest...

Join the game room at
Google Classroom to earn 5 exp
Class Code: **u7zumr36**

OVERVIEW OF DIGITAL IMAGE PROCESSING

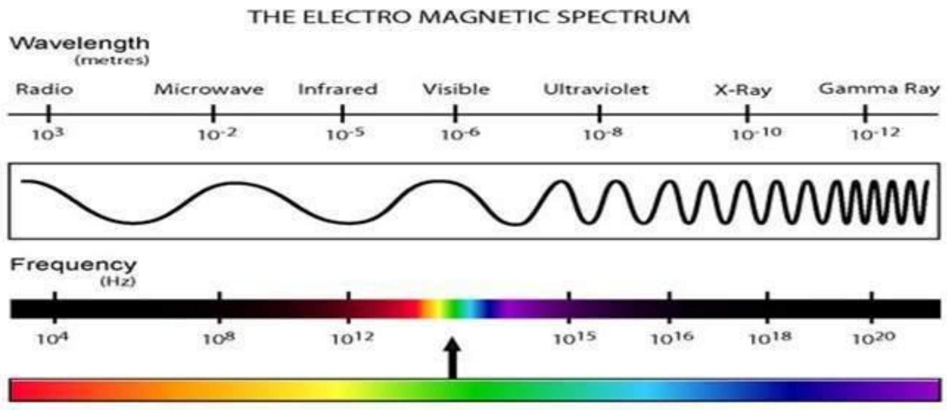
What is digital image processing?

- Digital image processing deals with manipulation of digital images through a digital computer
- “Image processing” is a subfield of signals and systems studies
- Digital image processing receives a digital image as the input, processes through algorithms, and give the output as a digital image



What is digital image processing?

- Analog Image is a 2-dimensional signal from sensors. To process an analog image using Analog Image Processing techniques, you must use the knowledge of electrical engineering



What is digital image processing?

- In our field, we focus on digital images which are already sampled from analog images

What is digital image processing?

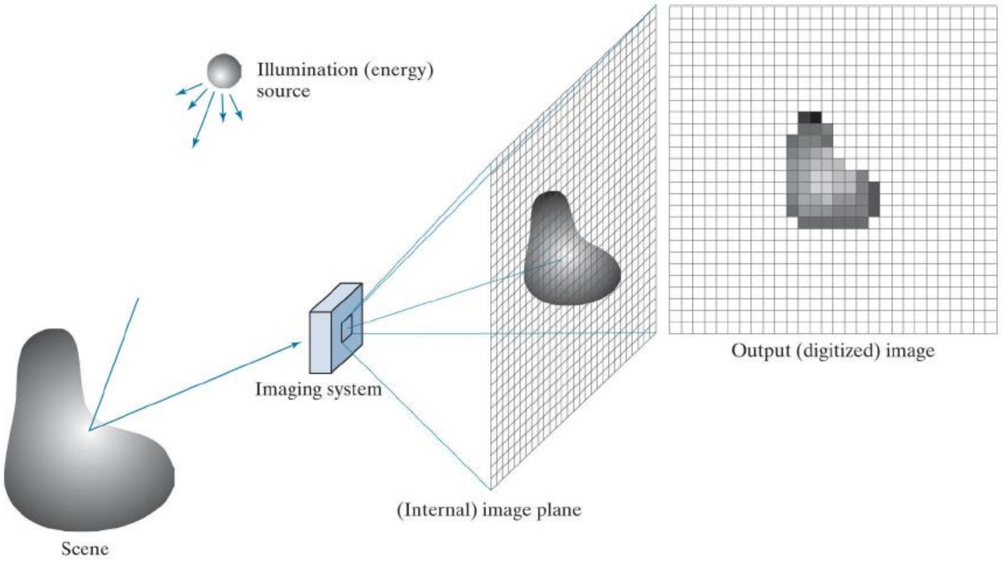
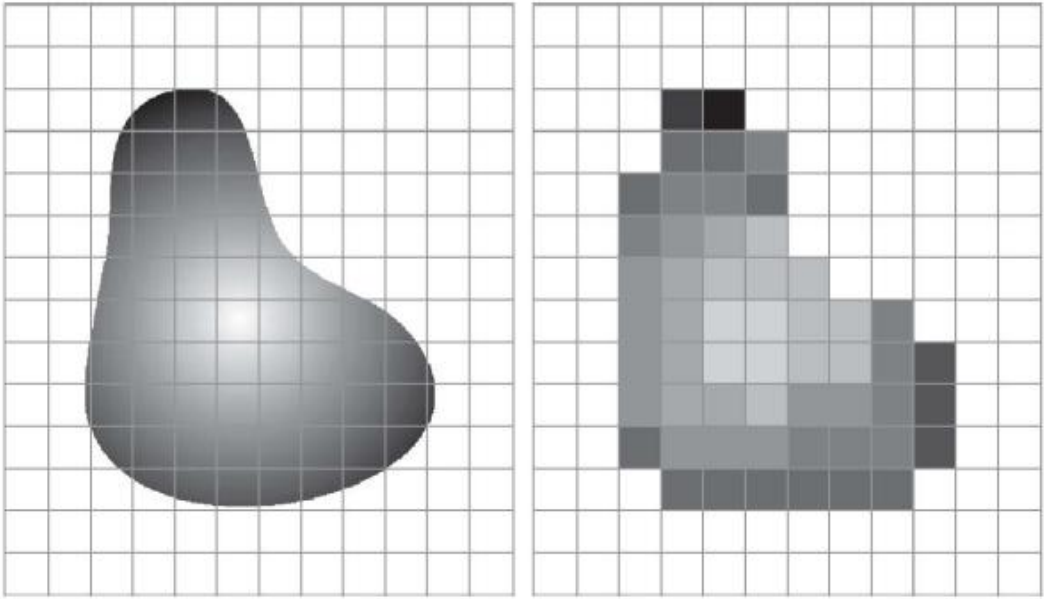


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

What is digital image processing?



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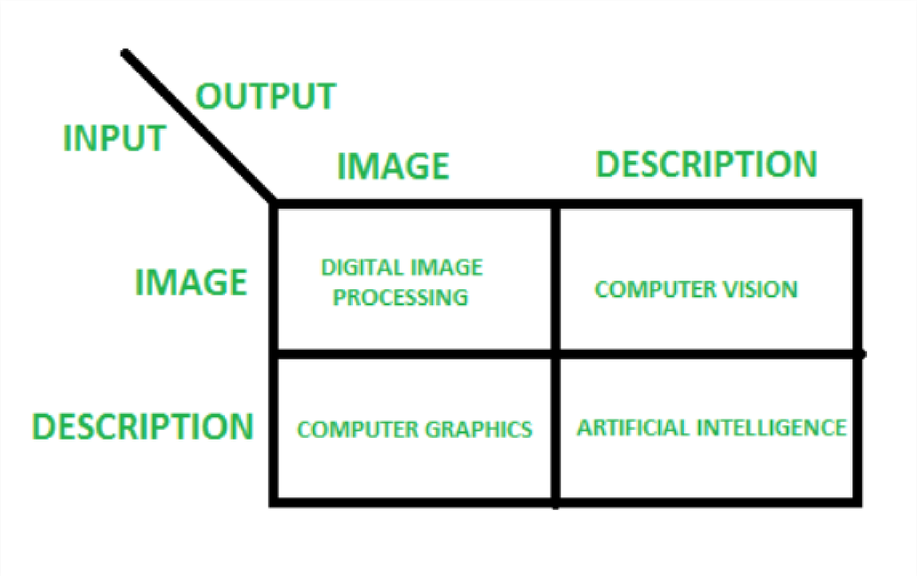
What is digital image processing?



Digital Image Processing

- Digital Image Processing is categorised into 3 groups
  - Low-level
  - Mid-level
  - High-level

Digital Image Processing

- Low-level processing
  - Involves primitive operations which results both input and output are images
  - Normally will be some pre-processing such as noise reduction, image enhancement, etc.

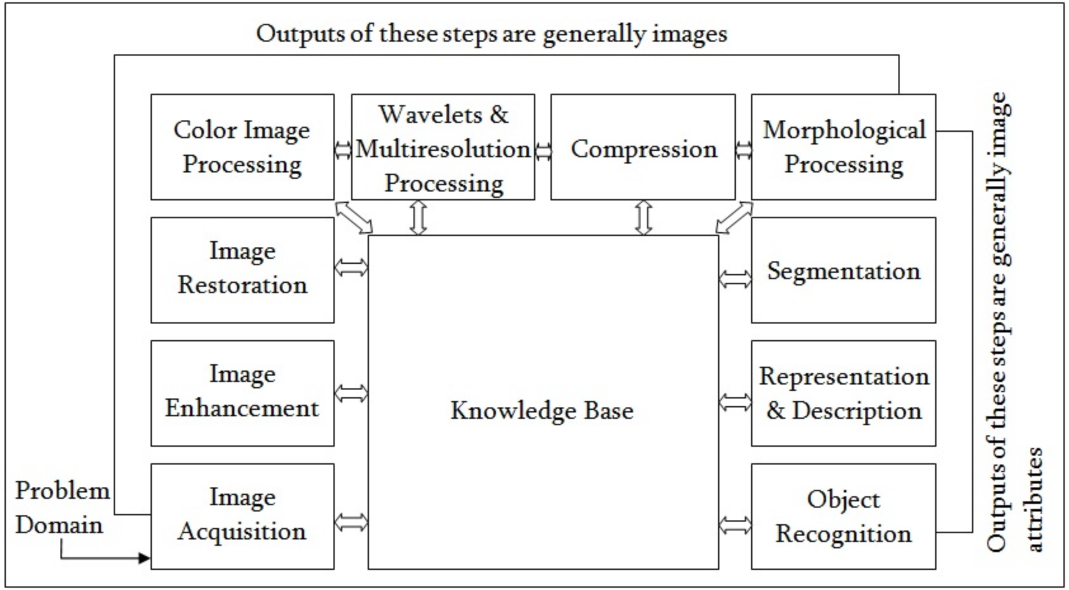
Digital Image Processing

- Mid-level processing
  - Involves more complex algorithm in order to extracts image attributes such as image segmentation, object recognition, edge detection, etc.

Digital Image Processing

- High-level processing
  - Involves converting images to something that make senses in the real world. Normally involve computer vision disciplines

Fundamental steps of digital image processing



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Prerequisites

- Calculus
- Probability
- Differential Equations
- Matrix
- Programming skills
  - We use Java OR C++ OR Python in this course!

Applications

- Image sharpening and restoration
- Medical field
- Remote sensing
- Transmission and encoding
- Machine/Robot vision
- Color processing
- Pattern recognition
- Video processing
- Microscopic Imaging
- Others

WHAT IS AN IMAGE

What is an image?

What is an image?

- The image on the right is a popular image for image processing studies called "Lena" or "Lenna"

What is an image?

- Image can be represented in many 2-dimensional form
  - Array
  - Matrix
  - etc

128	30	123
232	123	321
123	77	89
80	255	255

What is an image?

- An image is a two-dimensional signal, defined by function f(x, y) representing its coordinates and its intensity.

Image pixel at (m, n)
Denoted f(m, n)

What is an image?

- Location m and n is SPATIAL data
- Pixel value of each pixel f(m ,n) is INTENSITY data

Image pixel at (m, n)
Denoted f(m, n)

What is an image?

- Image has 3 types of resolution
  - SPATIAL Resolution => 640 x 480, 1024 x 768
    - Number of pixels in image area (frequencies of sampling signal)
  - TEMPORAL Resolution => 25 fps
    - Number of image for video frame in a second
  - BIT Resolution => 24 bit
    - Number of data in ONE pixel

IMAGE MODELS AND FILES FORMATS

Image colour model

Binary images

(Black and White)

- Binary images consist with only 2 colours: - black and white
- They use 1 bit to store each pixel of image

Image colour model

Grayscale images

- Grayscale images consist with 256 different levels of gray colour.
- However, human eyes can distinguish only about 40 levels of gray colour. So we can see them as a gradient.
- They use 8 bits to store the data of each pixel

Image colour model

RGB images

- Consist with a combination of 256 levels of each Red, Green, and Blue colours – which bring about 16.7 million different colours
- This is the number of colours which human eyes can distinguish, we refer this number as “True Colour”
- They use 24 bits to store the data of each pixel

Image colour model

HSV images

- HSV stands for HUE, SATURATION, and VALUE
- They are an alternative form of RGB. Hue represents colour (0 – 179), Saturation represents the white amount (0 – 255), and Value represents the black amount (0 – 255)
- HSV model is more often used in computer vision and image analysis due to its representation on grayscale values (S and V) which used a lot in image processing

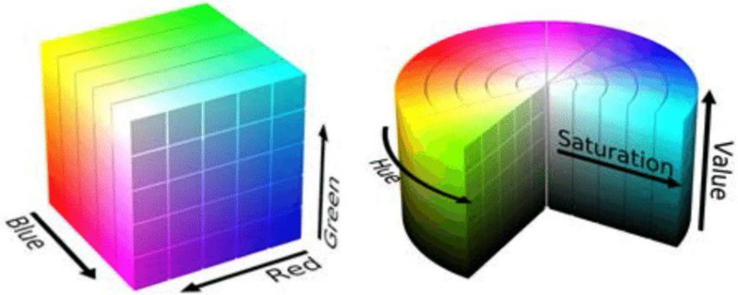
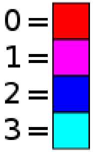


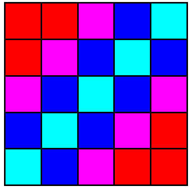
Image colour model

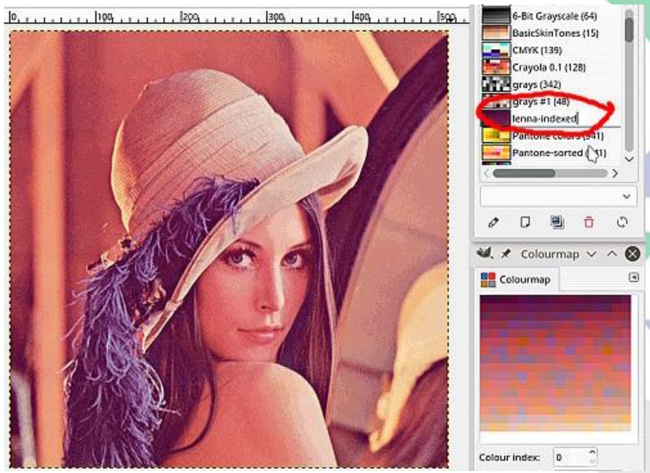
Indexed colour images

- Indexed colour is a model of image colour to save memory and storage
- The image data will not be stored in form of colour of pixels directly. Instead, it will be stored in colour data called “PALETTE” and each pixel will refer to the number of palette instead.

0	0	1	2	3
0	1	2	3	2
1	2	3	2	1
2	3	2	1	0
3	2	1	0	0







Examples of image file formats

- BMP (BitMap Pictures) – Lossless format
- GIF (Graphics Interchange Format) – Indexed format
- JPEG (Joint Photographic Experts Group) – Lossy format
- PNG (Portable Network Graphics) – open-source alternative of GIF
- TIFF (Tagged Image File Format) – Flexible format

BITMAP format

Bitmap format consists with 3 parts which is essential for digital image processing using C++

- Image Header (54 bytes)
- Colour Table (1024 bytes)
- Image Data (The rest of it)

10	4	offset to start of image data in bytes
14	4	size of BITMAPINFOHEADER structure, must be 40
18	4	image width in pixels
22	4	image height in pixels
26	2	number of planes in the image, must be 1
28	2	number of bits per pixel (1, 4, 8, or 24)
30	4	compression type (0=none, 1=RLE-8, 2=RLE-4)
34	4	size of image data in bytes (including padding)

End of Week 1